Customer Shopping Behavior Analysis

Exploratory Data Analysis & Business Intelligence Report

Project Scope: Customer Shopping Dataset Evaluation

Records: 3,900 Transactions

Tools: Python (pandas), SQLite / SQL, Power BI

Status: Final Deliverable Report

1. Executive Summary

This project examines the customer shopping dataset to identify spending patterns, product-category performance, customer groups, discount behavior, and subscription status. The workflow combines data preparation in Python, structured analysis in SQL, and visual reporting in Power BI. The dataset contains 3,900 purchase records and 16 standardized columns.

The analysis indicates that male customers generated higher total revenue than female customers in the analyzed dataset, middle-aged and adult cohorts drove the largest revenue shares among age groups, and customer segmentation identified a high proportion of repeat buyers.

2. Dataset Overview & Data Quality Verification

Attribute	Value / Status
Total Rows / Records	3,900 purchase transactions
Total Columns	16 original columns (standardized to snake_case) + engineered features
Missing Values	0 missing values across all columns
Duplicate Records	0 duplicate rows

3. SQL Analysis & Results

3.1 Revenue by Gender

Male customers generated substantially higher total revenue than female customers in this transaction set, driven by a higher volume of recorded purchases.

Gender	Total Revenue (USD)
Female	$75,191
Male	$157,890

3.2 Revenue by Age Group

Grouping customer ages into standard demographic bands demonstrates that middle-aged and adult cohorts drive the majority of retail revenue.

Age Group	Total Revenue (USD)	Revenue Share (%)
45–59 (Middle-Aged)	$67,711	29.05%
30–44 (Adult)	$64,927	27.86%
18–29 (Young Adult)	$57,279	24.57%
60+ (Senior)	$43,164	18.52%

3.3 Customer Segmentation

Customers were segmented based on their historical transaction count (`previous_purchases`):

Customer Segment	Definition Rule	Customer Count
Loyal	More than 5 previous purchases	3,476
Returning	1 to 5 previous purchases	424
New	0 previous purchases	0

4. Product Discount Analysis

The table below presents the top five products with the highest percentage of purchases utilizing a discount.

Product	Total Purchases	Discounted Purchases	Discount Percentage (%)
Hat	154	77	50.00%
Sneakers	145	72	49.66%
Coat	161	79	49.07%
Sweater	164	79	48.17%
Pants	171	81	47.37%

5. Strategic Recommendations

- **Targeted Loyalty Programs:** With over 89% of records classified under the Loyal segment (3,476 customers), retention offers should focus on tier-based perks rather than acquisition discounts.
- **Discount Policy Optimization:** High discount utilization on staples like hats and sneakers (approx. 50%) suggests margin dilution. A/B testing promotional thresholds is recommended.
- **Demographic Focus:** Marketing campaigns should prioritize Middle-Aged (45–59) and Adult (30–44) cohorts, who collectively account for nearly 57% of total revenue.